

Kritarth Dandapat

Incoming PhD Student, Michigan State University | Undergraduate Researcher, University at Buffalo

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Professional Summary

Incoming PhD student (Fall 2026) in the Department of Computer Science and Engineering at Michigan State University, joining the Data Mining Laboratory under Dr. Pang-Ning Tan, with a planned focus on **spatiotemporal machine learning**, **deep learning-based weather forecasting (DLWF)**, and **AI adversarial robustness**. Completing an accelerated three-year BS in Computer Science (Minor: Statistics) at the University at Buffalo, SUNY, with a 3.8+ GPA and Dean's List recognition every semester. Uniquely combines production-grade full-stack engineering (**React Native**, **YOLOv8** pipelines, **CUDA**-accelerated training) with advanced scientific ML (**E(3)-equivariant MPNNs**, symmetry-aware **GNNs**, **ALIGNN**) across mobile health AI and computational materials science. Co-author on manuscripts in digital health and materials discovery; recipient of the PEARL undergraduate research award (\$2,500) and the UB Health Futures Challenge second-place prize (\$1,000).

Education

PhD in Computer Science

Michigan State University

Starting Fall 2026

East Lansing, MI

- Department of Computer Science and Engineering; Data Mining Laboratory (Advisor: Dr. Pang-Ning Tan)
- Planned research focus: spatiotemporal machine learning, deep learning-based weather forecasting (DLWF), and AI adversarial robustness

Bachelor of Science in Computer Science, Minor in Statistics

University at Buffalo, SUNY

August 2023 – June 2026

Buffalo, NY

- Specialization in artificial intelligence; coursework spanning reinforcement learning, computer vision, machine learning, quantum computing, and distributed systems
- GPA 3.8+; Dean's List all semesters; accelerated three-year completion (21–22 credit semesters)
- Presidential Scholarship: \$15,000 per year; PEARL Award: \$2,500 competitive undergraduate research grant (Experiential Learning Network)

Research Experience

Research Assistant

Embedded Sensing and Computing (ESC) Group, University at Buffalo

Mobile sensing for healthcare and rehabilitation with Prof. Wenyao Xu.

June 2024 – Present

Buffalo, NY

OralScan: AI-Assisted Geriatric Oral Screening

June 2024 – February 2025

- Architected a full-stack mobile health platform (React Native, React web) integrating end-to-end **YOLOv8** vision pipelines for guided intraoral image acquisition, multi-class dental disease screening, and automated tooth numbering across temporal scan sessions.
- Formulated a spatio-temporal scan-guidance system that aggregates sequential intraoral frames to produce consistent disease-detection signals robust to user-induced motion variability.
- Co-authored formative usability and acceptability study submitted to *Smart Health* (under review); delivered oral presentations to clinical stakeholders at CTSI and SURC 2024.
- Awarded second place (\$1,000), University at Buffalo Health Futures Challenge (Spring 2026).

OrthoScan: Orthodontic Remote Monitoring

February 2025 – October 2025

- Engineered mobile and backend systems for at-home orthodontic progress tracking, deploying **YOLO**-based hardware detection paired with depth-assisted spatial measurement pipelines to quantify structural change

across longitudinal patient visits.

- Translated sequential intraoral imagery into calibrated geometric progress signals, enabling quantitative remote monitoring between in-office appointments and reducing reliance on subjective clinician assessment.

mRehab: Telerehabilitation Platform

November 2025 – Present

- Contributed to a remote rehabilitation platform incorporating session logging, multi-modal sensor ingest pipelines, time-series signal analysis, and exercise scoring algorithms deployed in live outpatient therapy workflows.

Undergraduate Researcher

Peng Research Lab, University at Buffalo

Computational materials science and clean-energy applications with Prof. Jiayu Peng.

June 2025 – Present

Buffalo, NY

Perovskite Ordering and Symmetry-Aware Graph Neural Networks

June 2025 – November 2025

- Benchmarked symmetry-aware **GNN** architectures (including **ALIGNN**) on ordering-dependent energetics in crystalline perovskites; contributed benchmarking results and revised analysis to arXiv:2409.13851 and the open-source **PerovskiteOrderingGCNNs** repository.
- Migrated experiment tracking infrastructure from SigOpt to **Weights & Biases**; automated training loops, hyperparameter sweeps, and multi-architecture evaluation pipelines, reducing manual overhead and improving reproducibility.
- Co-authored commentary on agentic AI for multimetallic catalyst discovery, published on *ChemRxiv* (2025).

Equivariant MLIPs, Alchemical Extensions, and Universal Atom Models

December 2025 – Present

- Implement **E(3)-equivariant** message-passing machine learning interatomic potentials (**MLIPs**) augmented with alchemical graph extensions to enable efficient composition sweeps and energy prediction over disordered crystalline structures.
- Integrate alchemical treatment with universal atom model (UMA) stacks (**MACE**) to accelerate structural relaxation and disorder characterization workflows in computational materials discovery pipelines.

Founder, Inference Foundry

Open Research Software Initiative

Starting May 2026

Remote

- Launched a collaborative open-source initiative building reproducible ML inference tooling, standardized research software scaffolding, and shared contributor infrastructure for the scientific ML community (inference-foundry.rweb.site).

Publications and Preprints

- Soni, P., **Dandapat, K.**, Gherardi, A., Bo, W., Li, R., and Xu, W. (2025). OralScan, an AI-Powered Mobile Tool for Geriatric Oral Healthcare: A Formative Usability and Acceptability Study. Submitted to *Smart Health* (under review).
- Peng, J., Liu, C., Luo, Y., and **Dandapat, K.** (2025). Accelerating Multimetallic Catalyst Discovery with Robotics and Agentic AI. *ChemRxiv*, version 1. DOI: 10.26434/chemrxiv-2025-13n3f.
- Peng, J., et al. (2024). Learning Ordering in Crystalline Materials with Symmetry-Aware Graph Neural Networks. *arXiv* preprint arXiv:2409.13851. (Contributed benchmarking and revisions.)

Research Presentations

- Oral presentation and live demo – CTSI Research Group of Doctors, Nurses, and Medical Students (April 2024)
- Oral presentation – SUNY Undergraduate Research Conference (SURC) (April 2024)

Selected Projects

PaddockPsychRL: F1-Informed Multi-Agent Reinforcement Learning *Spatiotemporal Modeling & Multi-A...*

- Formulated a multi-agent RL framework integrating Formula 1 temporal performance signals into agent psychology models within **PettingZoo** cooperative environments, mapping real-world behavioral dynamics to learned policy structure.
- Constructed driver-specific agent profiles from **FastF1** 2025 lap-time consistency metrics, encoding temporal performance variability as per-agent action-noise scales to reflect authentic decision-making under pressure.
- Benchmarked tabular Q-learning and **MAPPO** baselines against psych-conditioned policies; psych-aware agents improved Strategy Resilience Score from 0.54 to 0.63 (+16.7%) under noisy multi-agent coordination.

Stack: Python, FastF1, PettingZoo, Ray RLlib, MAPPO, Q-learning

Marine Guardian: Ship Detection in Satellite Imagery *Geospatial Computer Vision & Deep Learning*

- Designed a maritime monitoring pipeline grounded in geometric computer vision principles before applying deep models, achieving low-latency inference on high-resolution satellite imagery for spatial object detection.
- Engineered a roundness-based geometric classifier paired with **MobileNetV2**, attaining 98.72% ship-vs.-non-ship classification accuracy while eliminating heavy encoder overhead unsuitable for edge deployment.
- Benchmarked **EfficientNet** transfer learning with a custom decoder against ResNet-50 feature extraction and K-means clustering, systematically validating architecture decisions across spatial generalization scenarios.

Stack: Python, OpenCV, TensorFlow, EfficientNet, MobileNetV2, ResNet-50, K-means

People Counting via CSRNet Density Estimation *Deep Learning & Spatial Regression*

- Developed a density-estimation pipeline for crowd counting in highly occluded surveillance scenes where bounding-box detectors fail due to extreme pedestrian overlap, framing the task as spatial regression over density maps.
- Trained **CSRNet** end-to-end in **PyTorch** to predict per-pixel crowd density fields, enabling robust aggregate counts from integrated spatial density rather than fragile per-instance localization.
- Accelerated training and inference with **CUDA**; preprocessed surveillance-style frames with **OpenCV** for repeatable benchmarking across public crowd-counting datasets.

Stack: PyTorch, CSRNet, CUDA, OpenCV

Human Emotion Detection: Multi-Architecture Benchmark *Computer Vision & Model Robustness*

- Constructed a controlled multi-architecture benchmark comparing shallow CNN, **ResNet-34**, and **Vision Transformer (ViT)** backbones on facial expression recognition under a unified preprocessing and evaluation protocol, isolating architectural contribution from data effects.
- Optimized training schedules and augmentation strategies per architecture; ViT and ResNet-34 variants demonstrated superior generalization over the CNN baseline on held-out facial expression data.
- Achieved 87.5% top-1 classification accuracy on the best-performing configuration after systematic cross-architecture ablation.

Stack: PyTorch, TensorFlow, CNN, ResNet-34, Vision Transformer, OpenCV

Professional Experience

Teaching Assistant, CSE 341: Computer Architecture	January 2026 – May 2026
<i>Department of Computer Science and Engineering, University at Buffalo</i>	<i>Buffalo, NY</i>

- Delivered weekly VHDL recitations and office hours for 30+ students; graded exams and assignments; provided technical support to 100+ students on Piazza throughout the semester.
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Tutor and Peer-Assisted Leader	August 2024 – December 2025
<i>Tutoring and Academic Support Services, University at Buffalo</i>	<i>Buffalo, NY</i>

- Facilitated two interactive sessions per week in statistics and CS fundamentals; course feedback indicated approximately 30% improvement in quiz performance among attending students.
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Founder and Vice President

UB National Student Data Corps (NSDC)

October 2023 – May 2024

Buffalo, NY

- Co-founded the UB chapter; launched the chapter website, organized data science workshops, and built networking programs connecting undergraduates to research and industry opportunities.

Honors and Awards

- **PEARL Award** – \$2,500 competitive undergraduate research grant, University at Buffalo Experiential Learning Network (November 2025)
- **Health Futures Challenge** – Second place (\$1,000) for OralScan, University at Buffalo (Spring 2025)
- **Cybersecurity Excellence** – Top 100 globally, Northeastern Cybersecurity C2C Finals; 10th in world finals (2025)
- **Collegiate Lockdown** – Top two teams representing UB; 4th in finals (2025)
- **Dean's List** – University at Buffalo (Fall 2023, Spring 2024, Fall 2024)
- **Presidential Scholarship** – \$15,000 per year

Competitions and Challenges

- IAA AppXcelerate Application, University at Buffalo (January–March 2025) – Principal Investigator, OralScan
- Russell L. Agrusa CSE Student Innovation Competition (November 2024) – team member, OralScan
- Aging Innovations Challenge (November 2024) – team member, OralScan
- Community Champions for Disability Health Challenge (October 2024) – Principal Investigator, OralScan

Technical Skills

- **Languages:** Python, JavaScript, Java, C++, Rust
- **Machine learning and AI:** PyTorch, TensorFlow, computer vision, deep learning, graph neural networks (GNNs), YOLOv8, Vision Transformer (ViT), reinforcement learning (**MAPPO**, Q-learning), **Ray RLlib**
- **Scientific ML and materials:** **ALIGNN**, machine learning interatomic potentials (**MLIPs**), **MACE**, **E(3)-equivariant MPNNs**, **Weights & Biases**, PyTorch Geometric
- **Software engineering:** React, React Native, Node.js, Django, Git, **CUDA**, OpenCV, NumPy, Matplotlib
- **Databases and cloud:** MongoDB, SQL, Firebase, SQLite, Google Colab, Firebase Hosting

Research Interests

- Spatiotemporal machine learning and deep learning-based weather forecasting (DLWF)
- AI adversarial robustness and out-of-distribution generalization
- Geometric and equivariant deep learning; graph neural networks for structured data
- Computer vision and sequence modeling for healthcare and scientific applications
- Machine learning for materials science and scientific discovery

Certifications

- Machine Learning Specialization – Stanford University and DeepLearning.AI (Coursera)
- Deep Learning Specialization – Stanford University and DeepLearning.AI (Coursera)
- PyTorch for Deep Learning – Udemy
- Additional professional coursework: deep learning (TensorFlow), Django, React, Node.js, Python (Udemy)